

Research Internship Project Report

Surrogate Modeling in Statistical Learning: Theoretical Foundations, Algorithm Engineering, and Real-World Benchmark Applications of Gaussian Process Regression

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EXECUTIVE SUMMARY

During the 6-week research internship from **May 25, 2026, to July 5, 2026**, under the supervision of **Prof. Neelesh Shankar Upadhye**, I conducted an in-depth investigation into non-parametric Bayesian machine learning, specifically focusing on **Gaussian Process Regression (GPR)**. The primary achievements of this project include:

- Theoretical Foundations:** In-depth study of non-parametric surrogate modeling, weight-space vs. function-space views, inductive bias, likelihood formulations, spectral density (Bochner's Theorem), path differentiability, and kernel design based on *Gaussian Processes for Machine Learning (GPML)* (Chapters 1, 2, and 4).
- Algorithm Engineering (From Scratch):** Building robust numerical implementations of GPR and baseline regressors (Linear Regression, Polynomial Regression, KNN, Decision Trees) from first principles using **raw NumPy and SciPy**, incorporating **Cholesky matrix decomposition** ($LL^T = K + \sigma_n^2 I$) for numerical stability.
- Empirical Benchmarking:** Evaluating custom implementations against **Scikit-Learn** across 6 real-world engineering and environmental datasets (12 Jupyter Notebooks in total), demonstrating mathematically identical metrics.
- Presentation Slide Decks:** Creating two compiled Beamer slide decks for research communication.

Repository URL: https://github.com/Prateek312413/IITM_Internship

1 Theoretical Foundations & Mathematical Formulations

1.1 Supervised Learning & Inductive Bias

In supervised learning, given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ with $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$, the goal is to infer $y = f(\mathbf{x}) + \epsilon$.

- Under-determination Problem:** Training data alone cannot uniquely identify a function; infinitely many functions satisfy any finite dataset.
- Restriction Bias (Parametric View):** Constrains the solution space to a rigid parameter family $f(\mathbf{x}) = \mathbf{x}^T \mathbf{w}$, leading to under-fitting on non-linear manifolds.
- Preference Bias (Bayesian View):** Places a probability distribution over an infinite class of functions, preferring smoother curves while retaining flexibility.

1.2 Weight-Space Bayesian Linear Regression

Assuming a zero-mean Gaussian prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \Sigma_p)$ and i.i.d. Gaussian noise $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, the likelihood is $p(\mathbf{y} \mid X, \mathbf{w}) = \mathcal{N}(X\mathbf{w}, \sigma_n^2 I)$. Completing the square yields the Gaussian posterior over weights $p(\mathbf{w} \mid X, \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_{\text{post}}, \Sigma_{\text{post}})$:

$$\Sigma_{\text{post}} = \left(\Sigma_p^{-1} + \frac{1}{\sigma_n^2} X^T X \right)^{-1}, \quad \boldsymbol{\mu}_{\text{post}} = \frac{1}{\sigma_n^2} \Sigma_{\text{post}} X^T \mathbf{y} \quad (1)$$

Integrating over posterior weights yields the predictive distribution for a new input $\mathbf{x}_*$:

$$p(y_* \mid \mathbf{x}_*, \mathcal{D}) = \mathcal{N} \left(\frac{1}{\sigma_n^2} \mathbf{x}_*^T \Sigma_{\text{post}} X^T \mathbf{y}, \mathbf{x}_*^T \Sigma_{\text{post}} \mathbf{x}_* + \sigma_n^2 \right) \quad (2)$$

1.3 Feature Projections & The Dimension Bottleneck

Mapping inputs into an N -dimensional feature space $\phi(\mathbf{x}) \in \mathbb{R}^N$ requires inverting an $N \times N$ matrix $\Sigma_{\text{post}} = \left(\Sigma_p^{-1} + \frac{1}{\sigma_n^2} \Phi^T \Phi \right)^{-1}$. As $N \rightarrow \infty$, direct inversion becomes impossible.

Woodbury Matrix Identity Resolution: Applying $(B + UCV)^{-1} = B^{-1} - B^{-1}U(C^{-1} + VB^{-1}U)^{-1}VB^{-1}$, we rewrite the covariance as:

$$\Sigma_{\text{post}} = \Sigma_p - \Sigma_p \Phi^T (\Phi \Sigma_p \Phi^T + \sigma_n^2 I)^{-1} \Phi \Sigma_p \quad (3)$$

This transforms matrix inversion from feature size $N \times N$ to data size $n \times n$, defining the Kernel Trick $k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x})^T \Sigma_p \phi(\mathbf{x}')$ and establishing the **Function-Space View**.

1.4 Covariance Functions & Path Differentiability (GPML Chapter 4)

A Gaussian Process $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$ is specified by its mean and PSD kernel $k(\mathbf{x}, \mathbf{x}')$:

- **Bochner's Theorem:** A stationary kernel $k(\boldsymbol{\tau})$ is positive semidefinite iff it is the Fourier transform of a positive spectral density $S(\mathbf{s}) \geq 0$.
- **Differentiability:** A GP is m -times MS differentiable iff $\frac{\partial^{2m} k(\mathbf{x}, \mathbf{x}')}{\partial x_i^m \partial x_i'^m}$ exists and is finite at $\mathbf{x} = \mathbf{x}'$.
- **Kernel Families Studied:**
 - *Squared Exponential (RBF):* $k_{\text{SE}}(r) = \sigma_f^2 \exp\left(-\frac{r^2}{2\ell^2}\right)$ (Infinitely differentiable).
 - *Matérn Class:* $k_{\text{Matérn}}(r) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}r}{\ell}\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}r}{\ell}\right)$ ($\nu = 1/2$ OU process, $\nu = 3/2$ once differentiable, $\nu = 5/2$ twice differentiable).
 - *Rational Quadratic (RQ):* $k_{\text{RQ}}(r) = \sigma_f^2 \left(1 + \frac{r^2}{2\alpha\ell^2}\right)^{-\alpha}$ (Infinite scale-mixture of RBF kernels).

2 Algorithm Engineering & Implementation Details

Across 12 Jupyter Notebooks, two parallel framework pipelines were built:

1. From-Scratch Framework (NumPy/SciPy):

- Stable Cholesky decomposition $LL^T = K(X, X) + \sigma_n^2 I$.
- Triangular linear solves (`scipy.linalg.solve_triangular`) for $\boldsymbol{\alpha} = L^{-T}(L^{-1}\mathbf{y})$.
- Custom baseline regressors: `LinearRegressionFromScratch`, `PolynomialRegressionFromScratch`, `KNNRegressorFromScratch`, and `DecisionTreeRegressorFromScratch`.

2. Scikit-Learn Benchmarking Framework: Used `GaussianProcessRegressor` with L-BFGS-B hyperparameter optimization.

3 Real-World Benchmark Applications & Results

3.1 NASA Battery State-of-Health (SOH) Capacity Degradation

- **Dataset:** NASA B0005 Li-ion battery aging dataset (168 discharge cycles: 117 train, 51 test).
- **Results:** GPR (Matérn 5/2) achieved **RMSE = 0.010081 (98.872% Accuracy)**, outperforming Linear Regression (0.0166), Decision Tree (0.0430), and KNN (0.0455).

3.2 California Housing Value Prediction

- **Dataset:** StatLib California Housing dataset (20,640 total samples: 16,512 train, 4,128 test).
- **Results:** GPR (Rational Quadratic) achieved **RMSE = 0.552962 (78.907% Accuracy)**, outperforming KNN (0.6524), Decision Tree (0.7340), and Linear Regression (0.7451).

3.3 Delhi Air Quality Index (AQI) Forecasting

- **Dataset:** CPCB India database (16,000 hourly samples: 14,400 train, 1,600 test).
- **Results:** GPR (Matérn 3/2) achieved **RMSE = 37.366086 (88.055% Accuracy)**, outperforming KNN (42.945), Decision Tree (52.127), and Linear Regression (57.085).

3.4 Residential Building Energy Efficiency (ENB2012)

- **Dataset:** UCI Energy Efficiency dataset (768 samples: 460 train, 308 test).
- **Results:** Decision Tree achieved **RMSE = 1.0373 (97.113% Accuracy)**, while GPR (RBF+RQ Hybrid) achieved **RMSE = 1.3103 (94.982% Accuracy)**.

3.5 Gas Turbine NOx Emissions Prediction

- **Dataset:** UCI Gas Turbine dataset (7,151 cleaned samples: 5,720 train, 1,431 test).
- **Results:** GPR (Matérn) achieved **RMSE = 2.889904 (97.094% Accuracy)**, outperforming KNN (3.2411), Decision Tree (5.2496), and Linear Regression (6.1549).

3.6 SARCOS Anthropomorphic Robot Arm Dynamics

- **Dataset:** GPML SARCOS dataset (44,484 train / 4,449 test, sampled 5,000 train / 1,000 test).
- **Results:** GPR (Rational Quadratic) achieved **SMSE = 0.039985 ($R^2 = 0.960015$)**, outperforming KNN (0.0751) and Ridge Regression (0.0789).

4 Presentation Decks & Conclusion

Two Beamer slide decks were compiled and published:

1. `GPML_Chapters_1_and_2_Presentation.pdf` (13 slides: Intro, Inductive Bias, Weight Space, Woodbury Identity).
2. `GPML_Chapter_4_Presentation.pdf` (13 slides: Covariance Functions, PSD, Bochner's Theorem, Kernel Families).

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